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throughput of slotted aloha is double of pure aloha.

CSMA (carrier sense multiple access)
protocols
- is a network access method used on shared network topologies such as Ethernet to control access to network).

When energy is low → channel is idle.
When energy is moderate → channel is busy.
When energy is high → collision has occurred in network.

1-persistent CSMA
- station continuously sense the channel, till channel is busy
- once channel becomes idle, it will transmit immediately.
- If two station find channel idle at same time, then there is possibility of collision.

Non-persistent CSMA.
- if station finds a channel busy, it will wait for random amount of time and retransmission again.
- finding channel idle after waiting random amount of time at same time is less for different stations. Possibility of collision is less.

disadvantage → reduces efficiency of network because channel remains idle when there may be station with frame to send.

(iii) p - persistent CSMA

In p- persistent, one channel is idle it will transmit with probability p or it may not transmit with probability q. (1-p). If channel is busy, it will wait for next time slot.

- This method is used when channel has time slot duration ≥ max propogation delay time.

eg: There are 5 station in a slot. Probability of transmitting data is 0.4. If only one station should transmit. What is probability that one station will transmit

$^nC_r\, p^r q^{n-r}$ $\quad n=5,\ r=1,\ p=0.4$

$q = 1-p = 0.6$

$^5C_1 (0.4)^1 (0.6)^{5-1}$

$^5C_1 (0.4)\ (0.6)^4$

CSMA/CD → collision detection.

The place where collision happend is known as collision domain. When collision energy came to sender's interface, then collision is detected, sender will stop transmitting.

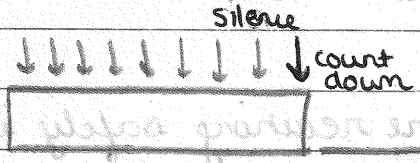
CSMA (fully discussed)

A station that wants to communicate 'listen' first on media communication and awaits a 'silence' of preset time. (called Distributed Interframe space or DIFS). After this compulsory period, station starts a countdown for random period considered. The max duration of countdown is called CW. (Collision window).

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If no equipment speaks before end of countdown, station simply transfer package. However, if it is overtaken by another station, it stops immediately its countdown and waits for next silence. Then it continues countdown from where it is stopped.

CSMA, was developed to overcome problem in ALOHA. ie minimise chances of collision.



CW (collision window)

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Error control policies.

error correction → Hamming code (XOR all combination and count no. of one's)

error detection → Parity — one bit error
CRC - cyclic redundancy check
checksum.

q: If probability of frame reaching safely is p, that means no. of transmission of frame is

$$= \frac{1}{1-(1-p)} = \frac{1}{p}$$

q: If probability of frame reaching safely is 0.1 then mean no. of transmission of frame is

$$\frac{1}{1-(1-0.1)} = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$$

i.e. 9 time failure, 10 time success.

(1) Parity (1 bit error)

data + parity bit = codeword.

codeword generated for data by circuit is known valid code word. & remaining all are invalid codeword.

A valid codeword can be converted to invalid codeword by noise (can be detected).

drawback: it cannot detect even no of errors.

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(II) Hamming distance.

performing XOR b/w two codeword will result in another codeword. The no. of one in resultant codeword is hamming distance. (minimum will be hamming code).

XOR (000, 011) XOR
XOR (011, 101) XOR
XOR (101, 110) XOR

000	011
011	101
011	110
= (2)	= (2)
hamming distance	hamming distance.

to detect d errors, minimum hamming distance = d+1
ex: 1 bit error, minimum hamming distance = (2)

to correct d errors, minimum hamming distance = 2d+1
1 bit error, minimum hamming distance = (3)

(III) Checksum.

Data + checksum = codeword.

235349 + 73 = 23534973

checksum = 23	recieved is 26504973
53	26
49	50
125	+49
↓	125
25	↓
+1	25
26	+1
9's complement ↓	26
99 − 26	99−26 = 73
= 73 checksum.	checksum

but transmitted bit were wrong, so this is vertical bit error.

If modified data is in such a way that vertically placed bits cancel each other effect, then calculated checksum will be equal to that of receiver checksum. Then such error cannot be detected. These error are known as "vertical bit error".

(iv) CRC

Data = 1100

CRC generated = x

$= 1 \times x^1 + 0 \times x^0$

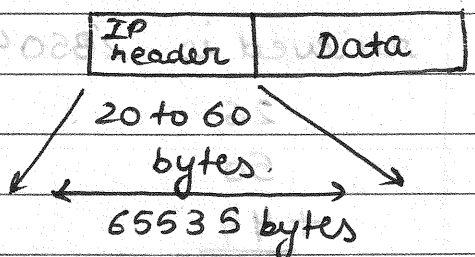
$= 10$

$$
\begin{array}{r|l}
10 & 1100\,0 \\
 & 10\downarrow \\
\hline
 & 10 \\
 & 10\downarrow \\
\hline
 & 000 \\
 & 00 \\
\hline
 & 0
\end{array}
$$

put resultant bit here.

Network layer:

dynamic header.



min = 20 bytes

max = 60 bytes.

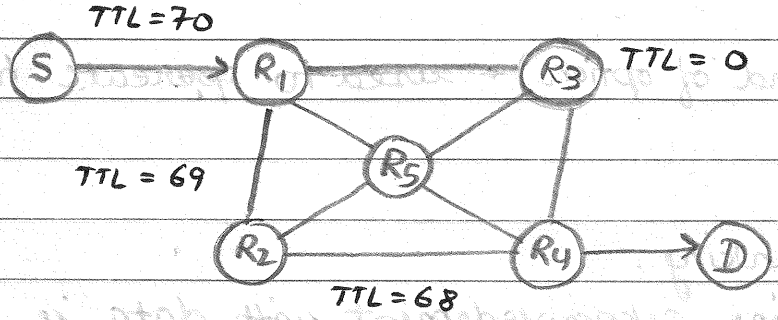
Source and destination address = 32 bits each

Data comes from transport layer.

Every LAN network will have its own MTU (maximum transferable unit). Fragments belonging to same packet will be given same identification no so that router can easily understand fragments belonging to which packet.

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TTL – time to leave field. (for identifying looping)



when a packet is forwarded to router the TTL value gets decremented. By chance if packet is in loop, then at one point of time TTL = 0, then next upcoming router will discard the packet.

Protocal field: it indicates type of application to which these packet belongs to that, as packet may be ftp, http, smtp etc.

checksum: is provided only for header (so that packet will be forwarded fastly).

source routing option → strict source routing
- path will be told by receiver to be followed (circuit switching)
- ex: S, R_1, R_2, R_4, R_6, D

↓

Loose source routing
- along with path specified by source + some other nodes are visited.

Routing option

(i) Time stamp option → roundtrip time b/w two end system.



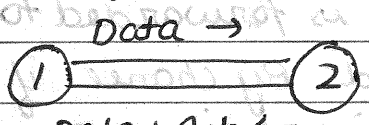
(ii) Record route option → used for debugging network & analysing network.

(iii) NOP option : used to fill gaps b/w option

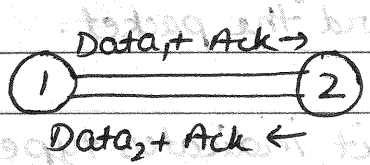
(iv) EOD (end of option) : used to seperate. header & data.

Piggy banking.
appending acknowledgment with data ie ACK + Data.

Half duplex :



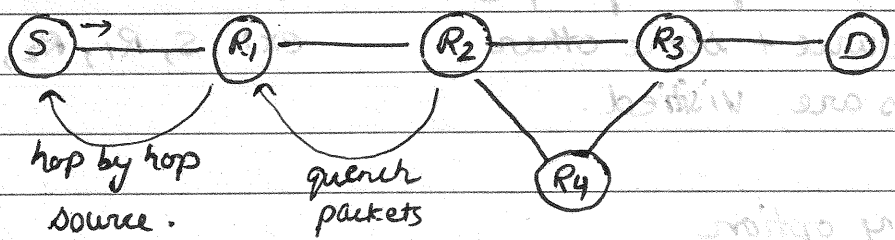
Full duplex :



no. of frames transmitted in Round trip time = window size.

ICMP. (reporting error & management queries)
Since IP has no error control. It depends on ICMP to provide error control. (it is used for reporting error & management queries).

(i) Source quench message.



- Wherever a router is congested then some of upcoming packets will be dropped.
- ICMP will take source IP from discarded packets & inform to source by sending source quench message.

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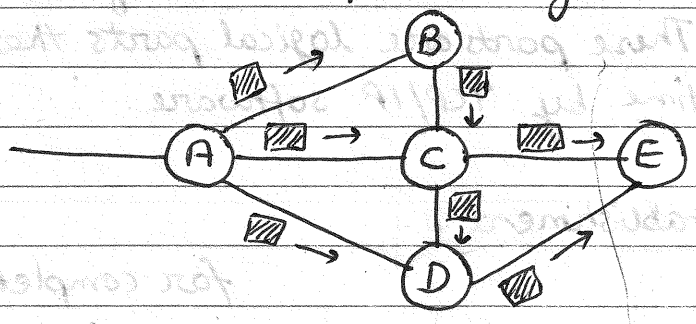
- source will reduce speed of transmission so that router will free from congestion.
- when congested router is far away from source, ICMP will send HOP by HOP source quench message so that every router will reduce speed of transmission via that path.

(11) Parameter problem.

Whenever a packet comes to router (then calculated header checksum should be equal to received header checksum) the only packet will be accepted by router else will be discarded. ICMP will take source address from discarded packet & informs source by sending parameter problem message.

- when some fragments are lost in network, holding fragments are dropped by router. ICMP informs source address.

Flooding (non adaptive algorithm)
is defined whenever a packet comes to router it is directed in all directions except point of origin. Flooding creates redundant packets by router.



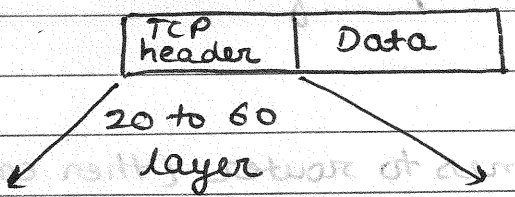
ABCE = 3 hops
ACE = 2 hops
ADCE = 3 hops

Flooding is used to find unknown destination that is IP address is known but physical location is not known.

Drawback - creates a redundant packet which may lead to congestion of router.

Transport layer. → TCP (segment)
→ UDP (datagram)

(1) TCP protocol. → works as segment.



source & destination address = 16 bits each

Port address = 0 to $2^{16}-1$

Max packet size = 0 to 65535

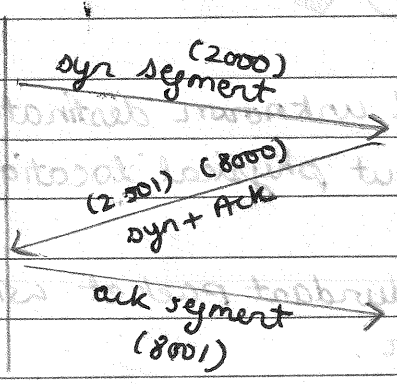
0 to 1023 → fixed port / universal port
predefined ports are those used for some predefined application – http, ftp, smtp etc.

1024 to 49151 → Registered port
are those ports which are used to test network software by companies.

49151 to 65535 → dynamic port, epimeral port
are those port which are used to distinguish seperately the process. These ports are logical ports that are given at run time by TCP/IP software.

(1) Connection establishment

for complete connection → 3 handshake is required.



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syn segment doesnot carry any data but it consumes seq no.

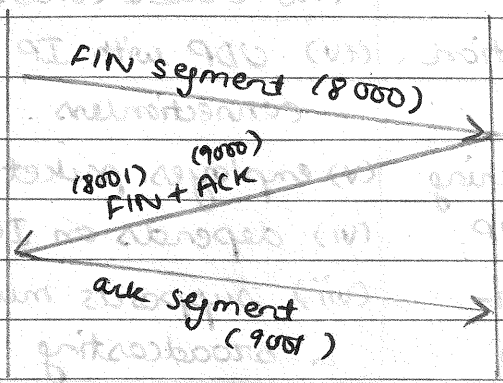
TCP support full duplex mode.

During connection establishment phase, we are transmitting control segments.

(ii) Data transfer phase.

flow control policies of transport layer. TCP can accept out of order segment but always send inorder Ack.

(iii) Connection release.



Whenever ack segment containing window size = 0 reaches client then persistent time is started.

Probe segment solves problem of deadlock (using persistent time).

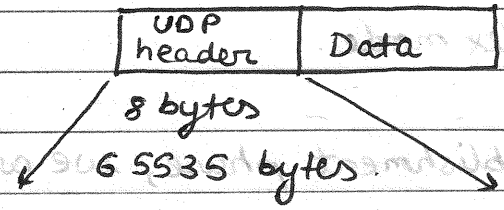
RST → is used to close connection suddenly & reseting again.

PSH segment → will never be lost (high priority)

URG → indicates urgent data. (high priority)

Urgent pointer → urgent data address will be available in this.

(II) UDP protocal: → works as datagram



source & destination addrees = 16 bits

Difference b/w TCP and UDP

	TCP	UDP
(I)	dynamic header	(I) static header (fixed)
(II)	provides flow control	(II) has no flow control
(III)	checksum is mandatory (has error control)	(III) checksum is optional (no error control)
(IV)	TCP with IP is connection oriented	(IV) UDP with IP is connectionless.
(V)	employes circuit switching	(V) employes packet switching
(VI)	doesnot depend on ICMP	(VI) depends on ICMP
(VII)	doesnot support multi-tasking & broadcasting	(VII) supports multitasking & broadcasting
(VIII)	slow	(VIII) fast

Congestion policies of TCP

Receiver's window will be known to sender during data transfer phase. This parameter will change for every Ack.

Private IP → 10.0.0.0 → 10.255.255.255
172.16.0.0 → 172.31.255.255
192.168.0.0 → 192.168.255.255
} works in LAN

free ↵

to avers globally, use Public IP
↳ not free.

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Classfull routing.

class A = 1
class B = 16
class C = 256
} private IP

(I) class A (raye 1 to 127)

Subnet mask 255.0.0.0 → 11111111.00000000.....
(8 | 8+8+8)
N/w | host

first bit {0}
ie $2^{24} - 2$ host
$2^{8-1} = 2^7$ network because {0} is fix as MSB.

(II) class B (raye 128 to 191)

Subnet mask 255.255.0.0
(8+8 | 8+8)
N/w | host

all classes have public & private IP address

168.202.32.4
↳ 1st octate compare raye class B.

first two bit {10}
ie $2^{16} - 2$ host
$2^{16-2} = 2^{14}$ N/w (network) because {10} is fixed as MSB

(III) class C (raye 192 to 223)

Subnet mask 255.255.255.0
(8+8+8 | 8)
N/w | host

first three bit {110}
ie $2^8 - 2$ host
$2^{24-3} = 2^{21}$ network because {110} is fixed as MSB.

(IV) class D → multicast address

(V) class E → reserved.

In host calculation, 2 IP address are decreased, they are not assigned because first IP is for network number and last IP is for Broadcast IP.

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class less interdomain routing (1)

↳ class A/8

↳ class B/16

↳ class C/24

169.254.0.0 - 169.254.0.16 → link local address

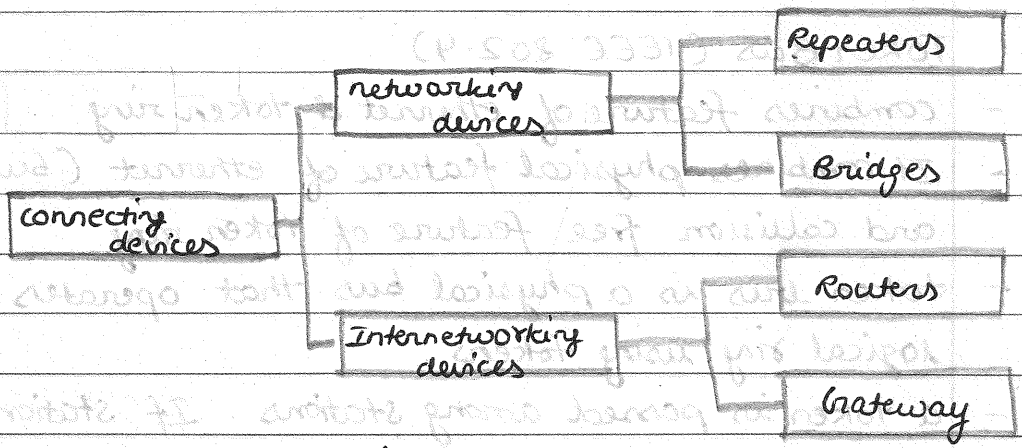
127.0.0.0 - 127.0.0.8 → loop-back address

0.0.0.0 - 0.0.0.8 → used to communicate over current network

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Inter network.

- When two or more networks are connected, then they become internetwork or internet.
- individual networks are joined into internetworks by use of internetworking devices.



- The interconnecting networking device, called routers and gateways are designed to overcome obstacles and interconnection without disrupting independent functioning of network.

Network Architecture.

- An open architecture allows system to be connected easily to devices and program made by other manufacturers.
- A closed architecture is one whose design is proprietary making it difficult to connect system to other system.

(i) Peer to Peer Architecture

- Each node has equivalent capabilities and responsibilities.
- simpler and less expensive
- donot offer same performance under heavy loads.

(ii) Client / Server Architecture

- Server will be powerful computer or processor.
- clients are less powerful PCs on which user run applications.

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Ethernet (IEEE 802.3)

- supports a LAN standard
- IEEE 802.3 defines two categories – baseband and broadband.
- CSMA/CD is access mechanism used in an ethernet.

Token Bus (IEEE 802.4)

- combines feature of ethernet & token ring
- It combines physical feature of ethernet (bus topology) and collision free feature of token ring
- token bus is a physical bus that operates as a logical ring using tokens.
- a token is passed among stations. If station wants to send data, it must wait and capture token.

Token Ring (IEEE 802.5)

- Each station must transmit during its turn and may send only one frame during each turn. The mechanism that coordinates this rotation is called token passing.
- a token is simple place holder frame that is passed from station to station around ring

FDDI (Fibre Distributed Data Interface) is LAN protocol.

- access method: token passing.
- access time is limited
- a station may send as many as frames as it can within its access allocated period.

Data link layer can be categorised by two types of layers → Medium access layer (MAL)
→ logical link layer

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HDLC protocal

- embeds information in data frame that allows devices to control data flow and correct errors.
- it is an ISO standard developed from SDLC (Synchronous Data Link control) standard proposed by IBM in 1970's.
- A session can use one of following connection modes, which determine how primary & secondary station interact.

(i) Normal unbalanced: secondary station responds only to primary station.

(ii) Asynchronous: secondary station can initiate a message.

(iii) Asynchronous balance: both stations send and receive over its part of duplex line. This mode is used in X.25 packet-switching network.

HDLC has three operation:

(i) NRM (normal response mode)
(ii) ARM (asynchronous response mode)
(iii) ABM (Asynchronous balanced mode).

HSSI is used to connect routers on Local area Network (LAN) with WAN (wide area network) over a high-speed line.

10-BaseT (cheapernet, thin ethernet, thinnet)

- uses thin coaxial cable.
- 10 Base 2 means 10Mbps speed and length 200m.

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Multiplexing

- set of techniques that allows simultaneous transmission of multiple signals across a single data link.

Types of Multiplexing.

(i) FDM (frequency division multiplexing)

- bandwidth of a link is greater than combined bandwidth of signals to be transmitted.
- channel must be seperated by strips of unused bandwidth (guard bands) to prevent signal from overlapping.

(ii) WDM (wave division)

- same as FDM, except light signal are transmitted through fibre optic channel. ie frequencies are very high.

(iii) TDM (time division)

- multiple transmission can occupy a single link by subdividing them and interleaving portion.
- can be synchronous TDM, Asynchronous TDM.

Trunk

- is a line or link designed to handle many signals simultaneously and that connects major switching center or nodes in communication system.
- are used to interconnect switches to form network and to interconnect LAN to form WAN or VLAN (Virtual LANs)
- can also be broadband wireless link
- use and management of trunks in a communication system is called trunking.

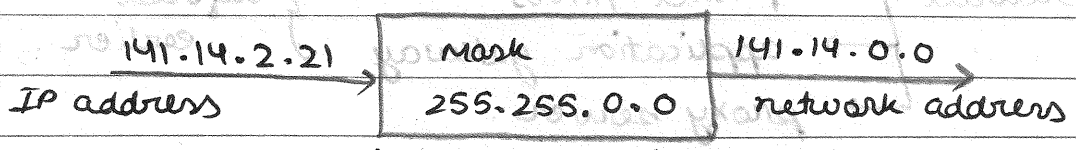
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Local Loops.

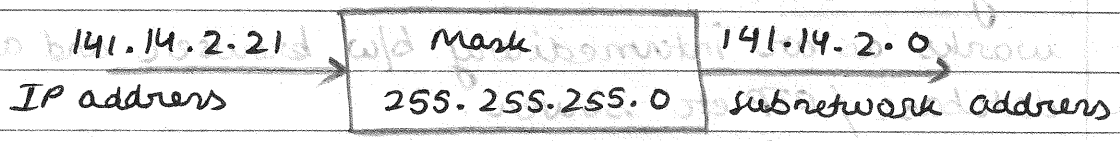
- is wired connection from telephone company's central office in a locality to its customer's telephones at homes & busines
- this connection is usually on pair of copper wires called twisted pair
- originally designed for voice transmission only using analog transmission on sigle voice channel.
- WLL is use of wireless communication link as "last mile / first mile' connection.

Masking (In classes subnetting)

- is a process that extracts address of physical network from an IP address (IP address is 32 bits)
- masking can be done whether we have subnetting or not.
- if we have not subnetted network, masking extracts network address from IP address.
- if we have subnetted, masking extracts subnetwork address from an IP address.



(a) without subnetting



(a) with subnetting.

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Other Protocols in Network layer.

(i) ARP (address resolution protocol)
- associated an IP address with physical address.
- used to find physical address of node when its internet address is known

(ii) RARP (reverse address resolution protocol)
- allows host to discover its internet address when physical address is known.

(iii) ICMP (Internet control message protocol)
- used by host and router to send notification of datagram problems back to sender.

(iv) IGMP (Internet Group message protocol)
- help a multicast router identify host in LAN that are members of multicast group.

Firewall → packet filters
→ application gateway } defined earlier.
→ proxy server.

Proxy server.
- works as an intermediary b/w browser and an database / FTP etc server.
- this proxy server translates b/w HTTP & FTP.
- keeps browser having to know many protocols.
- can cache previously requested pages.

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Congestion
When one part of subnet becomes overloaded (eg. one or more router), congestion results or when too many packets are present in subnet, performance degrades. This situation is called congestion.

DNS (domain name system)
↳ a system that can map a name to address and viceversa.

MAC (message authentication codes)
↳ a keyed function of message
- sender of message (m) computes MACk(m) and appends it to message
- Verification - receiver computes MACk(m) and compares it to received value.
- MAC keys are symmetric. Hence, doesnot provide non-repudiation
- MAC function doesnot need to be invertible.
- A MACed message is not necessarily encrypted.

How to ~~maintain~~ obtain (MAC) from block cipher.
(i) divide message into blocks
(ii) compute checksum by adding (XORing) them.
(iii) encrypt checksum.